Working With Lists

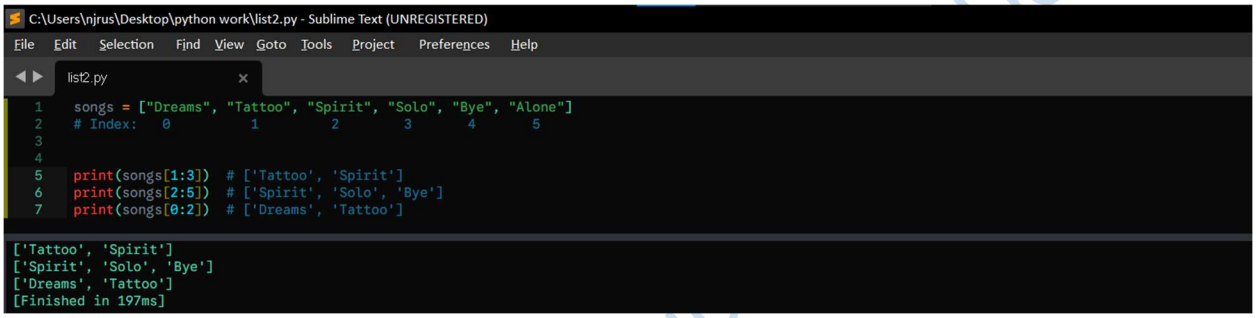
List Slicing

List slicing is a way of retrieving several items from a list by specifying a range of indexes. It is useful when you need a portion of a list rather than the whole list. This creates a copy of the list.

List indexing uses this syntax:

```
list_name[start:stop]
```

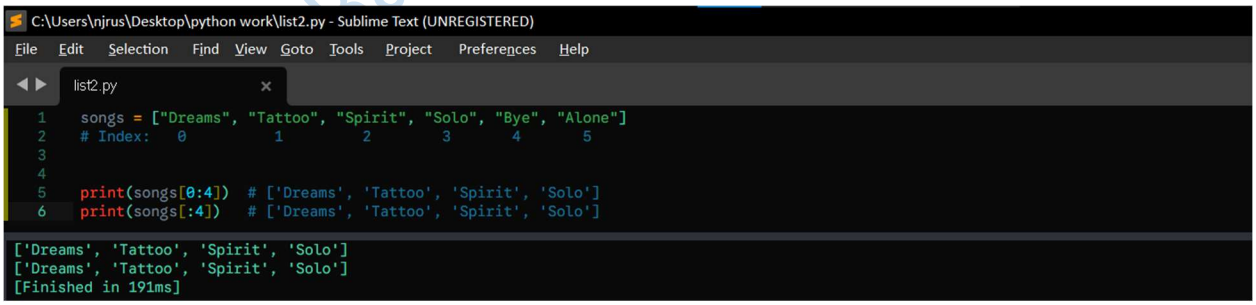
The **start** and **stop** represent **index values**. The **start index tells Python where to begin; the stop index tells Python where to stop.**



Note: Slicing includes the item at the starting index but excludes the item at the stop index. That is why song[1:3] includes songs at positions 1 & 2. Therefore, when creating copies of your lists using slicing, be careful how you supply your indexes.

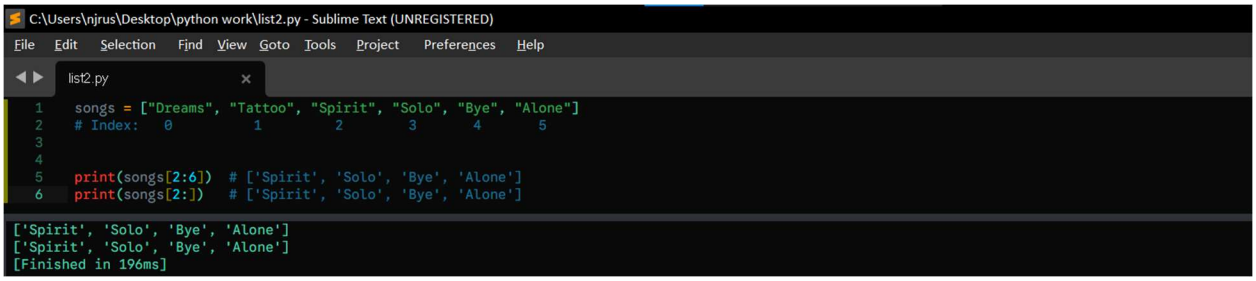
Omitting Start or Stop Indexes

However, in some situations, supplying the indexes may not be necessary. For example, if you need a copy of a list from, say the first item to the fourth item, you could either use `songs[0:4]` or `songs[:4]`. See below:



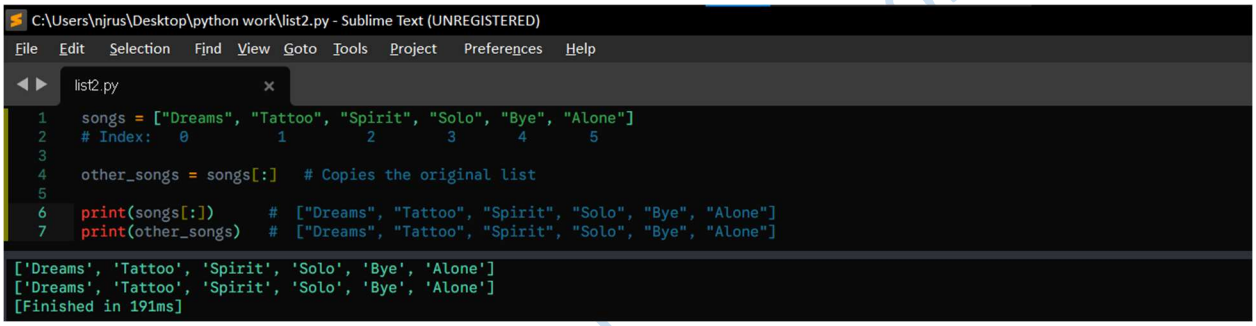
Both syntaxes imply the same thing and you'd still end up with the same copy.

If you need a copy, say from the second item to the end, you could either use `songs[2:6]` or `songs[2:]`. However, when you write larger programs, it might be difficult to track the index of the last item in your list. The syntax `songs[start_index :]` becomes handy in such situations.



Note: You may have noted that in line 5 I used index 6 as the stop index when there is no item at index 6 in songs list. However, remember that list slicing starts at the start_index you specify and does not include the stop_index. It stops at exactly stop_index – 1 and includes the item at stop_index – 1. This is why python does not raise an error, specifically IndexError.

You could also omit both the start and stop indexes when slicing. This creates a list including items from the start to the end.

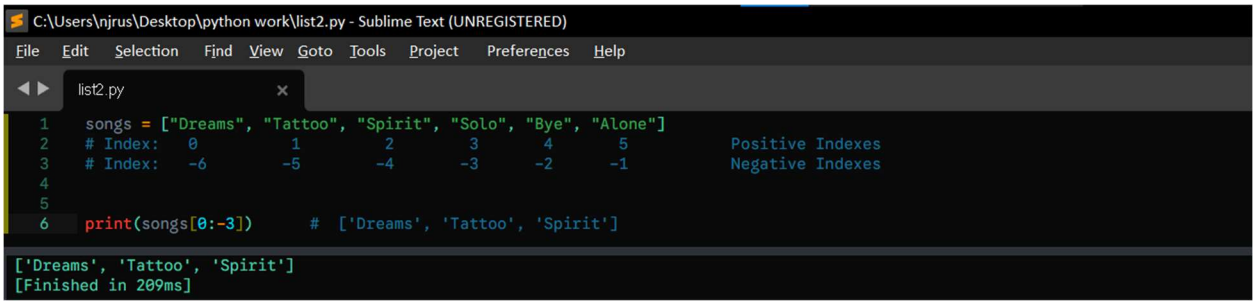


Note: This kind of slicing, i.e., original_list[:] is an alternative to the copy() method. Recall that the copy method is used to create a new list that has the same items as the items from the copied list. You could also use this slicing style and get yourseelf a copy of your original list.

Slicing with Negative Indexes

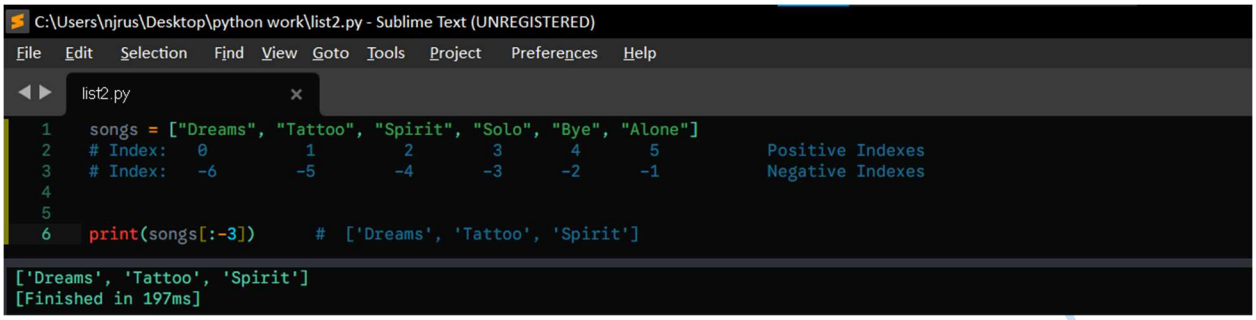
Slicing is not limited to positive indexes. You could use negative indexes and end up with the same list as you would while using positive indexes.

Recall that negative indexes count from the end of the list where -1 is the last item, -2 is the second-last, and so on.

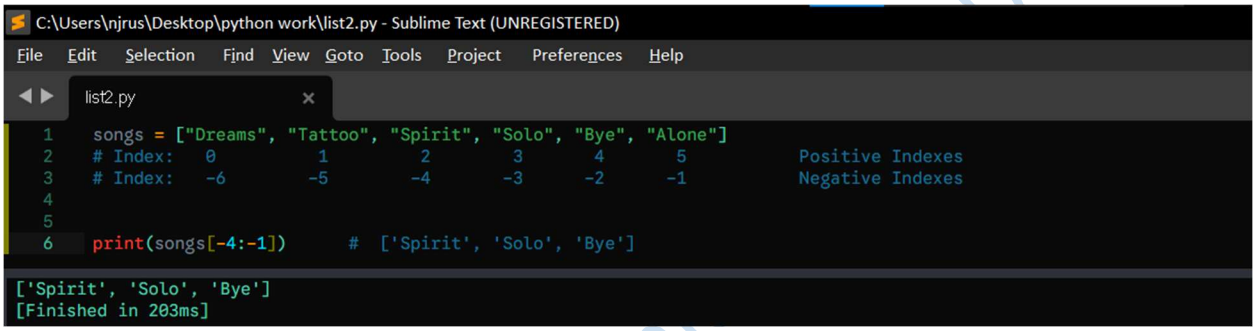


Here, slicing stops before the item that is at the third index from the end.

You can also ignore the starting or end positions.



This slices from the start to the index preceding index -3.

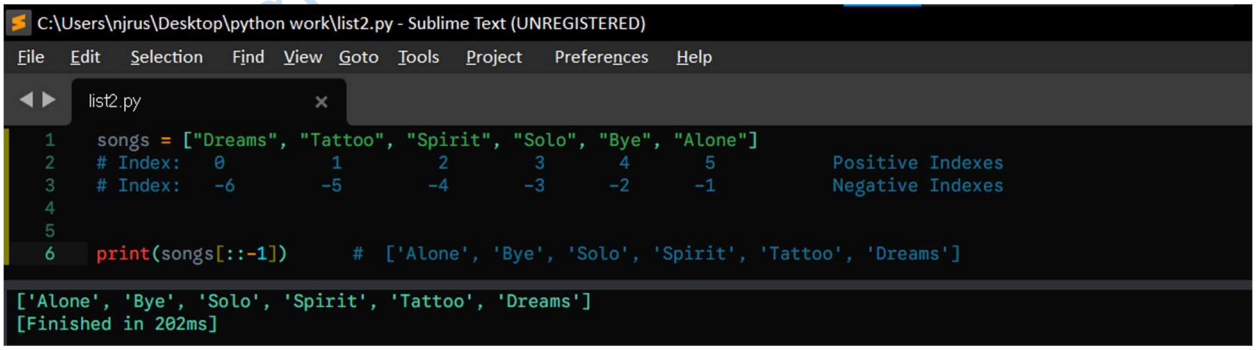


This slices from index -4 to -2.

Both positive and negative slicing work the same way. It is up to you to choose the one you are comfortable with and put practice on it.

Reversing a List with Slicing

You can also reverse the order of items in a list using slicing.



Recall the reverse() method which is used to reverse the order of items in a list. A disadvantage of using this method to reverse your lists is that once your list is reversed, the list is modified and will remain in that state while your program remains active.

Slicing avoids this by creating a copy such that your original list remains unmodified.



Note list_a got modified but list_b remained intact even after reversing its items with slicing.

I am, however not discouraging the use of reverse() method. There may be situations where its use is vital. So, assess the situation in your program before reversing your lists. That is, if you would still need the original list at some point later in your program, use slicing, if not, you can use reverse() or slice and work with the copy in your program.

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Try Yourself

A. Use this list for Exercises 1–5:

```
games = ["Chess", "FIFA", "Minecraft", "Roblox", "eFootball", "Asphalt"]
```

1. Write a slice that returns ["FIFA", "Minecraft", "Roblox"].

2. Write a slice that returns the first three items.

3. Write a slice that returns the last two items using negative indexing.

4. Write a slice that returns the list in reverse order.

5. What does games[2:5] return? Explain why the item at index 5 is not included.

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